

# Chapter 4

## Strings

# Character in C++

- The C++ library provides several macros for testing characters.
  - Be sure to include `ctype.h` header file

| Character Macro | Description                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| isalpha         | Returns true (a nonzero number) if the argument is a letter of the alphabet. Returns 0 if the argument is not a letter.             |
| isalnum         | Returns true (a nonzero number) if the argument is a letter of the alphabet or a digit. Otherwise it returns 0.                     |
| isdigit         | Returns true (a nonzero number) if the argument is a digit 0–9. Otherwise it returns 0.                                             |
| islower         | Returns true (a nonzero number) if the argument is a lowercase letter. Otherwise, it returns 0.                                     |
| isprint         | Returns true (a nonzero number) if the argument is a printable character (including a space). Returns 0 otherwise.                  |
| ispunct         | Returns true (a nonzero number) if the argument is a printable character other than a digit, letter, or space. Returns 0 otherwise. |
| isupper         | Returns true (a nonzero number) if the argument is an uppercase letter. Otherwise, it returns 0.                                    |
| isspace         | Returns true (a nonzero number) if the argument is a whitespace character. Whitespace characters are any of the following:          |

```
#include <iostream.h>
#include <ctype.h>

void main(void)
{
    char input;
    cout << "Enter any character: ";
    cin.get(input);
    cout << "The character you entered is: " << input <<
endl;
    cout << "Its ASCII code is: " << int(input) << endl;
    if (isalpha(input))
        cout << "That's an alphabetic character.\n";
    if (isdigit(input))
        cout << "That's a numeric digit.\n";
    if (islower(input))
        cout << "The letter you entered is lowercase.\n";
    if (isupper(input))
        cout << "The letter you entered is uppercase.\n";
    if (isspace(input))
        cout << "That's a whitespace character.\n";
}
```

# ***Program Output With Example input***

Enter any character: A [Enter]

The character you entered is: A

Its ASCII code is: 65

That's an alphabetic character.

The letter you entered is uppercase.

## ***Program Output With Other Example input***

Enter any character: 7 [Enter]

The character you entered is: 7

Its ASCII code is: 55

That's a numeric digit.

# *Strings*

- A **string** is a sequence of characters (letters, numbers, symbols) under double quote.
  - **String s=“Hello”**
- **C++** has no built-in String type like some programming languages.
- Strings are implemented as null terminated(`\0`) arrays of characters.
  - `char abc [3] = {'m','e'}` or `char abc [3] = “me”`
  - array abc consists of “m”, “e”, null
- `abc = “me”` (shortcut for defining and initializing a string)
- **Two types** of strings in C++:
  - **C-style string** → array of characters (old way)
  - **std::string** → sequence of characters modern, easier, recommended

# Review of the Internal Storage of C++ strings

- A C++ string is a sequence of characters stored in consecutive memory locations, terminated by a null character.

e.g. `char string[7] = "Bailey";`

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| B | a | i | l | e | y | \0 |
|---|---|---|---|---|---|----|

# Library Functions for Working with strings in C++

- The C++ library has numerous functions for handling C-strings. These functions perform various tests and manipulations.
- string functions are in **<string.h>**
- So we need to include the string.h header file to use the Functions.
  - String str="hello"
    - str.length(); or str.size() // output 5
  - A[5]={'H','e','l','l','o'};
    - sizeof(A) / sizeof(A[0])
    - Char str[]="Hello";
      - strlen(str)
  - }
- Add (concatenate ) strings using + operator



| Function | Description                                                                                                                                                                                                                                                                                                                                                                             |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| strlen   | Accepts a C-string or a pointer to a string as an argument. Returns the length of the string (not including the null terminator. <i>Example Usage:</i> len = strlen(name);                                                                                                                                                                                                              |
| strcat   | Accepts two C-strings or pointers to two strings as arguments. The function appends the contents of the second string to the first string. (The first string is altered, the second string is left unchanged.) <i>Example Usage:</i> strcat(string1, string2);                                                                                                                          |
| strcpy   | Accepts two C-strings or pointers to two strings as arguments. The function copies the second string to the first string. The second string is left unchanged. <i>Example Usage:</i> strcpy(string1, string2);                                                                                                                                                                          |
| strncpy  | Accepts two C-strings or pointers to two strings and an integer argument. The third argument, an integer, indicates how many characters to copy from the second string to the first string. If the <i>string2</i> has fewer than n characters, <i>string1</i> is padded with '\0' characters. <i>Example Usage:</i> strncpy(string1, string2, n);                                       |
| strcmp   | Accepts two C-strings or pointers to two string arguments. If <i>string1</i> and <i>string2</i> are the same, this function returns 0. If <i>string2</i> is alphabetically greater than <i>string1</i> , it returns a negative number. If <i>string2</i> is alphabetically less than <i>string1</i> , it returns a positive number. <i>Example Usage:</i> if (strcmp(string1, string2)) |
| strstr   | Accepts two C-strings or pointers to two C-strings as arguments, searches for the first occurrence of <i>string2</i> in <i>string1</i> . If an occurrence of <i>string2</i> is found, the function returns a pointer to it. Otherwise, it returns a NULL pointer (address 0). <i>Example Usage:</i> cout << strstr(string1, string2);                                                   |

## C++ std::string style functions

- `string s1 = "Hello";`
- `string s2 = "world";`
- **Copy:** Assignment = or `assign()` :
  - `string s2;`
  - `s2 = s1;` or `s2.assign(s1); //copy`
- **Comparing:** `==`, `!=`, `<`, `>`, or `.compare()`
  - `if (s1 == s2) cout << "Equal\n";`
  - `if (s1 != s2) cout << "Not equal\n";`
  - `int res = s1.compare(s2); //// Using compare()`
- **Concatenating:** `+` operator or `.append()`
  - `string s3 = s1 + s2; // Hello, World`
  - `s1.append(s2);`

Before the call to `strcat (String1, String2)`:

String1

|   |   |   |   |   |  |    |  |  |  |  |  |  |
|---|---|---|---|---|--|----|--|--|--|--|--|--|
| H | e | l | l | o |  | \0 |  |  |  |  |  |  |
|---|---|---|---|---|--|----|--|--|--|--|--|--|

String2

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| W | o | r | l | d | ! | \0 |
|---|---|---|---|---|---|----|

After the call to `strcat (String1, String2)`:

String1

|   |   |   |   |   |  |   |   |   |   |   |   |    |
|---|---|---|---|---|--|---|---|---|---|---|---|----|
| H | e | l | l | o |  | W | o | r | l | d | ! | \0 |
|---|---|---|---|---|--|---|---|---|---|---|---|----|

String2

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| W | o | r | l | d | ! | \0 |
|---|---|---|---|---|---|----|

```
#include <iostream.h>
```

```
#include <string.h>
```

```
void main() {
```

```
    char str1[] = "String test";
```

```
    char str2[] = "Hello";
```

```
    char one[10];
```

```
    strncpy(one, str1, 9);
```

```
    one[9] = '\0';
```

```
    cout << one << endl;
```

```
    strncpy(one, str2, 2);
```

```
    cout << one << endl;
```

```
    strcpy(one, str2);
```

```
    cout << one << endl;
```

```
}
```

# String compare

```
#include <iostream.h>
```

```
#include <string.h>
```

```
void main() {
```

```
    cout << strcmp("abc", "def") << endl;
```

```
    cout << strcmp("def", "abc") << endl;
```

```
    cout << strcmp("abc", "abc") << endl;
```

```
    cout << strcmp("abc", "abcdef") << endl;
```

```
    cout << strncmp("abc", "ABC",2) << endl;
```

```
}
```

# String Output

- A string is output by sending it to an output stream
- E.g.  

```
char str[20]="hello";  
cout<<" the string str is:"<<str;
```

# String input

- When the input stream **cin** is used space characters, **newline**, **tab**, etc. are used as separators and terminators.
  - skips over any leading spaces
  - terminates reading a value when it finds a white-space character
  - E.g.

```
Char name[20];  
cout<<"enter your name";  
cin>>name
```

Input: My name is Dawit  
Output: My

```
// This program cycles through a character array,  
    displaying  
// each element until a null terminator is encountered.  
#include <iostream.h>  
  
void main(void)  
{  
    char line[80];  
    int count = 0;  
  
    cout << "Enter a sentence of no more than 79  
characters:\n";  
    cin.getline(line, 80);  
    cout << "The sentence you entered is:\n";  
    while (line[count] != '\0')  
    {  
        cout << line[count];  
        count++;  
    }  
}
```



# ***Program Output with Example input***

Enter a sentence of no more than 79 characters:

**C++ is challenging but fun! [Enter]**

The sentence you entered is:

C++ is challenging but fun!

# Exercise

- Write a program to display characters of a string one by one.

```
#include <iostream>
using namespace std;
int main() {
    char str[] = "Hello World";
    int i=0;
    cout << "String = " << str << endl;
    cout << "Print characters:\n";
    while (str[i] != '\0')
    {
        cout << str[i];
        i++;
    }
    return 0;
• }
```